



Do Quang Hao

Computer Science Graduate Student

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Research Profile

Final-year Computer Science student at Ho Chi Minh University of Technology (GPA: 3.6/4.0) with interests in computer vision, natural language processing, multimodal learning, graph machine learning, and efficient AI systems. Experience includes industry-linked face recognition and edge vision, empirical benchmarking for image, text, and multimodal classification, and research on scene-graph-augmented visual question answering. Strongest fit is research-oriented ML work that values careful evaluation, reproducible implementation, and clear empirical reporting.

Education

Bachelor's Degree - Computer Science - Ho Chi Minh University of Technology 10/2022 - 07/2026
GPA: 3.6/4.0

Technical Skills

English	IELTS 7.0
Programming Languages	Python, C++, C, Haskell, Kotlin, JavaScript
ML / DL Stack	PyTorch, YOLO, TensorRT, ONNX, TensorFlow Lite, FAISS, LLMs
Systems / Backend	Docker, FastAPI, WebSockets, Redis, PostgreSQL, SQLite
Research Areas	Computer Vision, NLP, Multimodal Learning, Graph Neural Networks, Time Series, Speech Processing
Workflow / Tools	Git, Shell, LaTeX, Postman

Projects

Open-Set Face Recognition with Embedding Retrieval and Similarity Search 09/2023 - 12/2023
Industry Collaboration - IVS Company

- Studied an open-set face recognition pipeline using InsightFace and FaceNet embeddings with FAISS-based nearest-neighbor retrieval and PostgreSQL-backed metadata storage.
- Structured enrollment and search endpoints with FastAPI to support repeatable experiments on gallery construction, similarity search, and response latency.
- Containerized the system with Docker and documented engineering trade-offs for deployment-oriented experimentation.

Scene-Graph-Augmented Visual Question Answering with LLMs 02/2024 - 05/2024
Research Project with Assoc. Prof. Dr. Quan Thanh Tho - Ho Chi Minh University of Technology

- Investigated a visual question answering pipeline that combines scene-graph knowledge with a large language model for more grounded reasoning over image content.
- Built an interface for image upload, question answering, and response visualization to support iterative analysis of image-question pairs.
- Evaluated answer relevance, qualitative reasoning quality, and inference efficiency while documenting limitations such as detection errors and data complexity.

Empirical Sentiment Classification on Sentiment140 01/2025 - 05/2025
Machine Learning Assignment - Ho Chi Minh University of Technology

- Compared multiple sentiment classification baselines on the Sentiment140 dataset under a unified preprocessing and evaluation pipeline.
- Studied tokenization, feature extraction, and fine-tuning strategies while tracking accuracy, F1, and representative error patterns.
- Summarized empirical findings in report form and discussed trade-offs across model families.

MiniGo Language Design <i>Compiler / Language Design Project</i>		01/2025 - 05/2025
- Built a compiler-style pipeline covering grammar analysis, semantic error checking, and translation of MiniGo programs to JVM bytecode. - Used the project to strengthen understanding of language design, static analysis, and execution flow on the Java Virtual Machine.		
CV Builder Web <i>Frontend Web Project - HTML, CSS, JavaScript</i>		01/2025 - 05/2025
- Developed a template-driven CV editor with form-based input, template selection, and PDF export. - Focused on usability and layout control for document-style web interfaces.		
Bookstore Web <i>Frontend Web Project - HTML, CSS, JavaScript</i>		01/2025 - 05/2025
- Implemented a web application for buying and selling books with a user-oriented browsing and interaction flow. - Practiced client-side interface design, state handling, and feature integration in a multi-page web project.		
Image Classification: CNNs vs Vision Transformers <i>Deep Learning and Its Applications (CO3133) - Ho Chi Minh University of Technology</i>		03/2026 - 05/2026
- Built preprocessing, augmentation, training, and evaluation pipelines for multi-class image classification. - Fine-tuned pretrained CNN and Vision Transformer backbones under matched data splits and hyperparameter settings for fair comparison. - Reported accuracy, macro-F1, confusion patterns, and qualitative misclassifications to analyze architectural trade-offs.		
Text Classification: RNN / LSTM vs Transformers <i>Deep Learning and Its Applications (CO3133) - Ho Chi Minh University of Technology</i>		03/2026 - 05/2026
- Prepared text normalization, tokenization, batching, and training workflows for multi-class text classification. - Compared recurrent baselines such as RNN / LSTM with Transformer-based models under consistent evaluation settings. - Analyzed accuracy, macro-F1, and representative failure cases to study where recurrent and attention-based models differ.		
Research Experience		
Quantum Data Encoding for Quantum Machine Learning <i>Computer Science Thesis with Assoc. Prof. Dr. Thoai Nam and Dr. Diep Thanh Dang - Ho Chi Minh University of Technology</i>		06/2024 - 06/2026
- Reviewing literature and experimental directions at the intersection of quantum computing and quantum machine learning. - Formulating an undergraduate thesis topic around how quantum methods may be applied to machine-learning workflows.		
Experience and Activities		
Collaborator at IVS		11/2023 - 02/2024
Worked in an industry-linked setting connected to face-recognition experimentation and deployment-oriented computer vision.		
AI VIET NAM - AIO2024		2024
Participated in structured AI learning activities during the AIO2024 program.		
Internship at Codepresso		05/2025 - 08/2025
Completed a summer internship and gained exposure to professional engineering workflow and collaboration.		
Honors, Competition, and Training		
University Merit-based Scholarship		12/2023
Academic Incentive Scholarship		08/2022
ICPC Regional Ha Noi		12/2024